

Sensors for monitoring signs of pregnancy

EDV28A-N-01Specification



EDV28A-N-01



Product Features

- Utilising cutting-edge 60G millimetre-wave sensing technology, it detects millimetre-scale micro-movements of the chest in real time, providing comfortable, imperceptible, non-contact and long-term stable monitoring.;
- By combining edge computing with intelligent algorithms, the system continuously monitors and analyses the subject's presence, as well as key vital signs such as heart rate, respiration and body movement;
- It intelligently monitors sleep stages such as light sleep, deep sleep, REM sleep and wakefulness, enabling users to assess and review their sleep quality;
- Intelligently detects abnormal breathing or heart rate, with support for timely alerts, safeguarding users' health and safety;
- There are no privacy risks associated with the camera; privacy regulations are strictly adhered to at both the physical and data layers, ensuring the security of user information;

Electrical specifications

Operating voltage of the unit ^①	5~24V DC
Power consumption	Average power consumption < 0.4 W, maximum power consumption < 1 W
Contact details	WEBSOCKET

Performance Specifications

Detection range	<1 m, default 0.7 m
On/off detection time	Typically, it remains in position for 1–2 seconds and moves away for 5–45 seconds.
Measurement range	Respiratory rate measurement range: 12–40 bpm Heart rate measurement range: 60–120 bpm
Reporting cycle	1–60 S, default 3 S

Notes:

① UDP, MQTT and HTTP interfaces can be customised as required.

Launch parameters	
Radar transmission frequency	59~64GHz
Radar transmit power	≤15dBm (max)

Operating parameters	
Operating temperature	-20~60℃
Storage temperature	-20~85℃

Functional Specifications

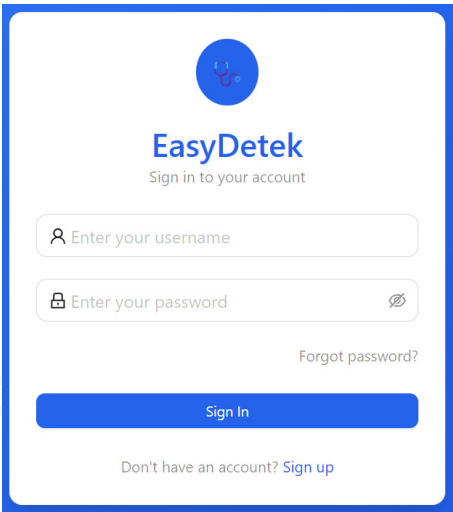
1.List of Features:

Reports the following statuses in real time and supports statistical analysis

Sleep statistics and analysis	Real-time physiological parameter values
Bedtime	Away/Sleep mode
Wake-up time	Physical activity scores
Duration of sleep	Heart rate
Sleep Quality Score	Respiratory rate
Proportion of light sleep/deep sleep/REM sleep	Heart rate status
Number of turns	Respiratory rate status

Function Description

View vital signs data via the web interface:
Go to the webpage: <https://easydetekiot.com/login>
Enter your username and password to log in.



Once logged in, simply tap the device card for the relevant sensor to view your real-time heart rate,real-time respiratory rate, real-time active/inactive status and sleep status, historical heart rate and respiratory rate trend graphs, and sleep quality reports.



1.Real-time physiological parameter values:

Present/Absent status: When a person enters the detection zone, 'Present' is displayed; after the person has been absent for a certain period, 'Absent' is displayed.

Wakefulness/Sleep States: Once the sensor detects that a person is asleep, it will identify the stages of REM sleep, light sleep and deep sleep, and switch between them accordingly.

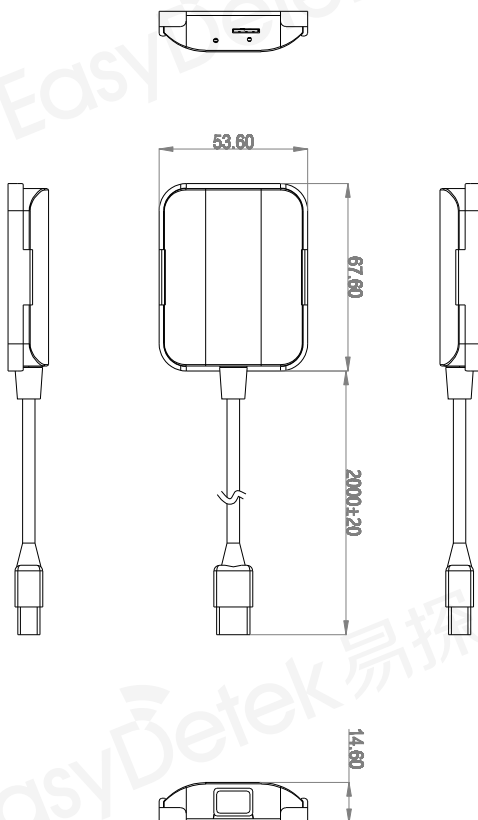
Respiratory and heart rate readings: When a person enters the monitoring area, their respiration and heart rate readings are not displayed immediately; these values are only displayed once the subject has reached a state of rest.

2.Vital signs and sleep statistics

Once the user is at rest, vital signs can be recorded in as little as 10 seconds; if the user is not at rest, the sensor may be unable to record valid vital signs until the user is at rest.

Product dimensions

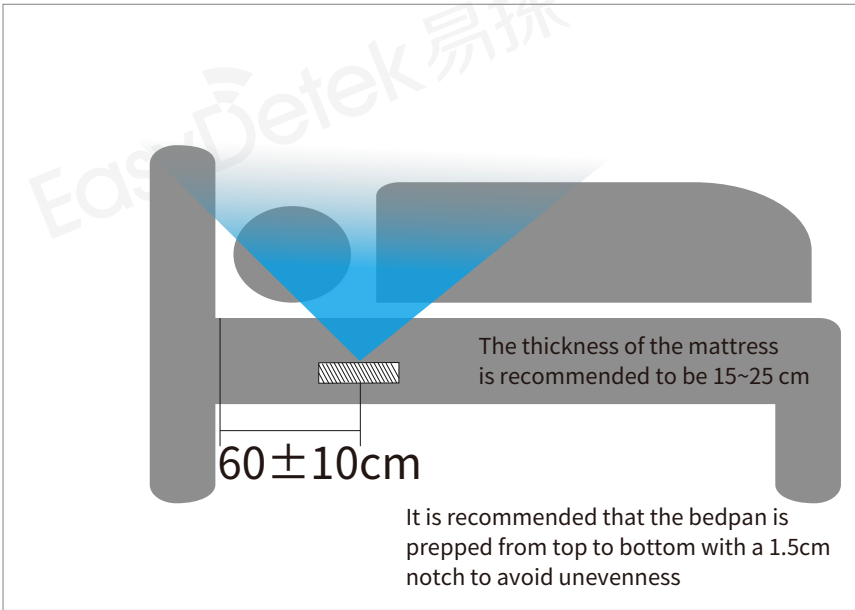
Units of measurement: mm



 Status Indicator

The green indicator light is constantly on	Power-on initialisation is in progress
The green indicator light flashes once	Submitting physiological monitoring data
The red indicator light is constantly on	Network connection failed
The red indicator light flashes once	Send a device online communication heartbeat
The yellow indicator light is flashing continuously	An OTA update is in progress

 Installation location



Installation location:

The sensor should be placed beneath the mattress and above the bed base; when the person is lying flat, their chest should be directly above the sensor.

We recommend that the mattress be between 15 and 25 cm thick.

Product Naming Rules

ED	Frequency Band	Major Product Category	Product Subcategory	Serial Feature Code	Function Extension	Running Number
ED	V	2	8	A	N	01
EasyDetek	C 5.8GHz	1 Microwave module	0 Ultra-low-power series	0-9,A-Z	Y With light sensor	A Matter 1 Cat 1
	X 10.5GHz	2 Standalone sensor	1 Flagship series		N No light sensor	C Casambi 2 2.4G
	Q 24GHz	3 Antenna / component	2 Side-mounted series		G Dry contact	D Dali 4 RS-485
	V 60GHz	4 MCU	3 External adjustable series		S AC switch	K KNX B Bluetooth
	S Ecological Products	5 Accessory	5 General-purpose series			M Mijia P PLC
		6 IC	6 Outdoor		T Tuya	R RF433
		7 AI sensor	7 TBD			W WiFi
		8 Networking	8 Basic series			Z Zigbee
		9 TBD	9 High-ceiling series			

Configuration Version Description

【Part number】: EDV28A-N-01
【PCB version number】: EDV28A_VB
【Software version number】: EDV28A V206

History of Revisions

Version	Time	Description	Notes
V1.0	2026-07-06	First edition	-

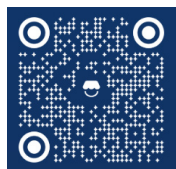
Points to Note

1. The radar should be installed in a location where it is not affected by nearby sources of vibration, including but not limited to: refrigerators, air-conditioning outdoor units, water pumps and generators.
2. Within the radar's detection range, avoid placing objects that sway, move or vibrate noticeably, as well as large metal surfaces, including but not limited to: pendant lights, fans, air conditioning units, houseplants, curtains and mirrors with a metallic coating.
3. When installing the radar sensor, if there is an aluminium substrate or other metal plate behind it, the sensor should be raised to a certain height, maintaining a distance of at least 5 mm from the metal surface; it must not be placed directly against or in contact with the metal surface. There must be no metal obstructions or high-current cables covering the front of the radar sensor's antenna; the sensor must not be positioned directly opposite the drive power supply, and should be kept as far away as possible from high-power components such as the rectifier bridge, transformer and switching transistors in the drive power supply.
4. Radar sensors penetrate plastic and wooden materials effectively, but cannot penetrate metal or materials with a metallic coating. If the casing of your product is made of special materials such as glass, ceramic or carbon fibre, please refer to the actual test results. If necessary, please contact a technical representative from Yitan Technology to carry out suitability testing.
5. Excessive power supply ripple may interfere with the radar sensor and cause false alarms; it is recommended that the power supply ripple be kept below 100 mV.
6. The presence detection function works by using radar to detect the movement of the chest and abdomen as a person breathes; consequently, false negatives (reporting no one is present when someone is actually there) or false positives (reporting someone is present when no one is there) may occur in the following situations:
 - The human body is in a non-resting state
 - The body is in a posture or position that makes detection difficult
 - The sensor is positioned or angled in a way that hinders detection
 - There are several people in adjacent seats
 - The body is covered by excessively thick duvets or clothing
 - A mattress that is too thick or too thin
 - A mattress that is too firm or too soft
7. To achieve better presence detection and enhance the user experience, we recommend the following:
 - Ensure that there are no moving objects within the detection area, including, but not limited to, fluttering curtains, swaying plants and rotating fans;
8. If the radar sensor is used in conjunction with wireless communication modules (such as NB, Bluetooth, Wi-Fi or 2.4 GHz modules), a sufficient distance should be maintained between them. It is recommended to keep a distance of at least 1 metre from high-power wireless communication devices, such as routers and wireless hotspots, during installation.
9. Yitan Technology is committed to providing customers with high-quality radar sensors that deliver an enhanced user experience. Product versions may be updated or revised without prior notice. Should you require further information, please contact our sales team to obtain the latest product documentation.

 For further information, please visit



WeChat Official Account



WeChat Shop